CAPSTONE PROJECT | RETAIL ANALYTICS

Retail Intelligence

Sales, Profit & Inventory Optimization

A Data-Driven Strategy for Sustainable Retail Growth

Sector	Retail / E-Commerce Analytics
Team Members	Alice Johnson · Raj Patel · Priya Sharma · Mohammed Ali
Faculty Mentor	Dr. Sunita Verma, Dept. of Computer Science
GitHub	github.com/your-org/retail-analytics-capstone
Tableau	public.tableau.com/your-dashboard-link

Context & Problem Statement

BUSINESS CONTEXT

- The retail sector faces intense margin pressure as discount-driven growth erodes profitability across all channels.
- With 673,000+ transactions across 4 years (2019–2023), the business generated $1.59B in revenue across 3 categories, 4 customer segments, and 3 shipping channels.
- Despite strong top-line revenue, strategic blind spots persist: which cities, products, and segments truly drive sustainable profit?
- Management lacks a unified, data-driven framework to allocate resources, optimize inventory, and prioritize high-value channels.

CORE BUSINESS PROBLEM

"How can management increase sales, profit margin, and stock efficiency by identifying the best-performing cities, categories, channels, brands, and customer segments?"

DECISION MAKER & OBJECTIVES

- **Decision Maker: VP of Retail Operations & Supply Chain Head**
- Identify top revenue & profit-generating cities, segments, and product lines
- Pinpoint underperforming channels and prescribe corrective actions
- Optimize inventory turnover to reduce carrying costs and stockouts
- Deliver an interactive Tableau dashboard for real-time operational decisions

Data Engineering & ETL Pipeline

673,502	26	4 Years	$1.59B
Total Records	Feature Columns	2019 – 2023	Total Revenue

ETL PIPELINE

01 EXTRACT

Raw CSV ingested from transactional POS system — 673K rows, 26 columns spanning 2019–2023. Dataset includes geography, customer, product, pricing, shipping, and profit variables.

02 TRANSFORM

Removed 422 anomalous rows (region='40.2'); Filled missing customer_age values with segment median; Parsed order_date & ship_date to datetime; Derived shipping_days, order_year, order_month, and age group buckets.

03 LOAD

Cleaned dataset exported for Python EDA (Pandas/Seaborn/Matplotlib) and loaded into Tableau Public for interactive executive and operational dashboards.

KEY COLUMNS USED IN ANALYSIS

city / state / region	Geographic performance mapping
customer_segment	Segment-level profitability & behavior
product_category / sub_category	Category margin & volume analysis
sales	Total revenue per transaction
profit	Net profit after costs & discounts
product_base_margin	Baseline profitability indicator
discount	Price reduction impact on margin
ship_mode	Channel (delivery) analysis
order_quantity / unit_price	Inventory & pricing efficiency
shipping_cost / shipping_days	Logistics cost & speed analysis

KPI & Metrics Framework

TOTAL REVENUE

$1.59B

Why it matters:

Measures the absolute business scale and tracks top-line growth year-over-year.

AVG BASE PROFIT MARGIN

51.3%

Why it matters:

Core profitability indicator — drives all resource allocation decisions.

AVG DISCOUNT RATE

12.5%

Why it matters:

Excessive discounting erodes margin; monitoring reveals pricing discipline gaps.

AVG SHIPPING TIME

~5.5 days

Why it matters:

Logistics efficiency KPI — directly impacts customer satisfaction & carrying cost.

CUSTOMER SEGMENTS

4 Segs

Why it matters:

Segment revenue & margin reveal where to focus acquisition & retention efforts.

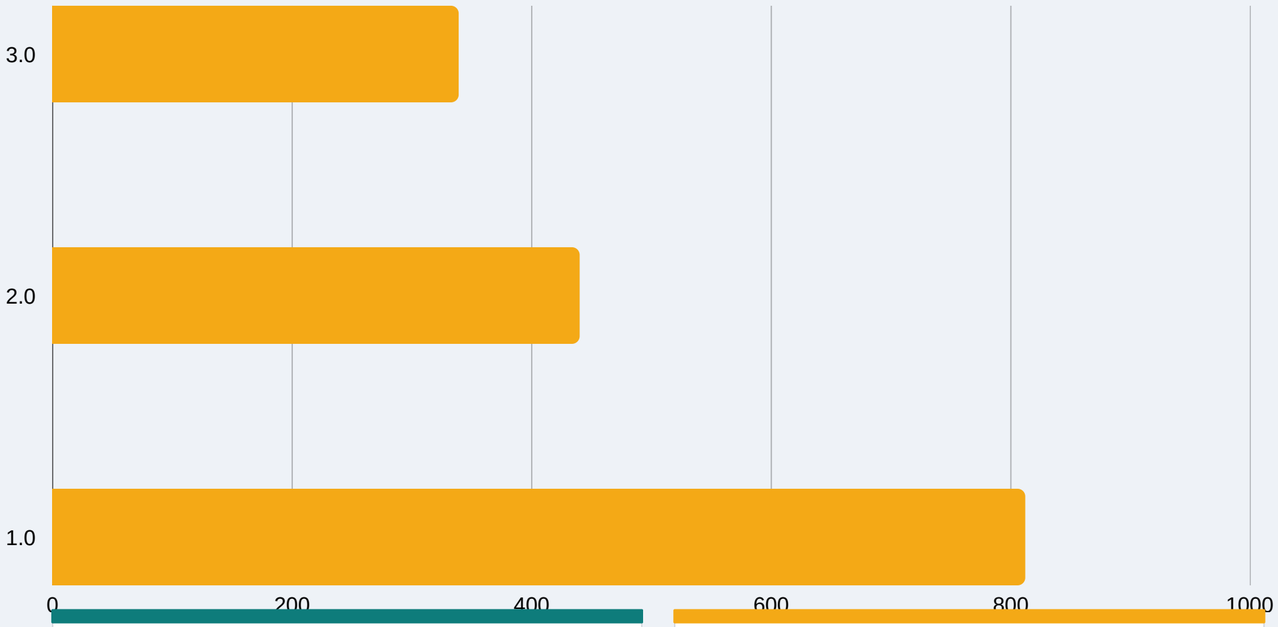
SHIPPING CHANNELS

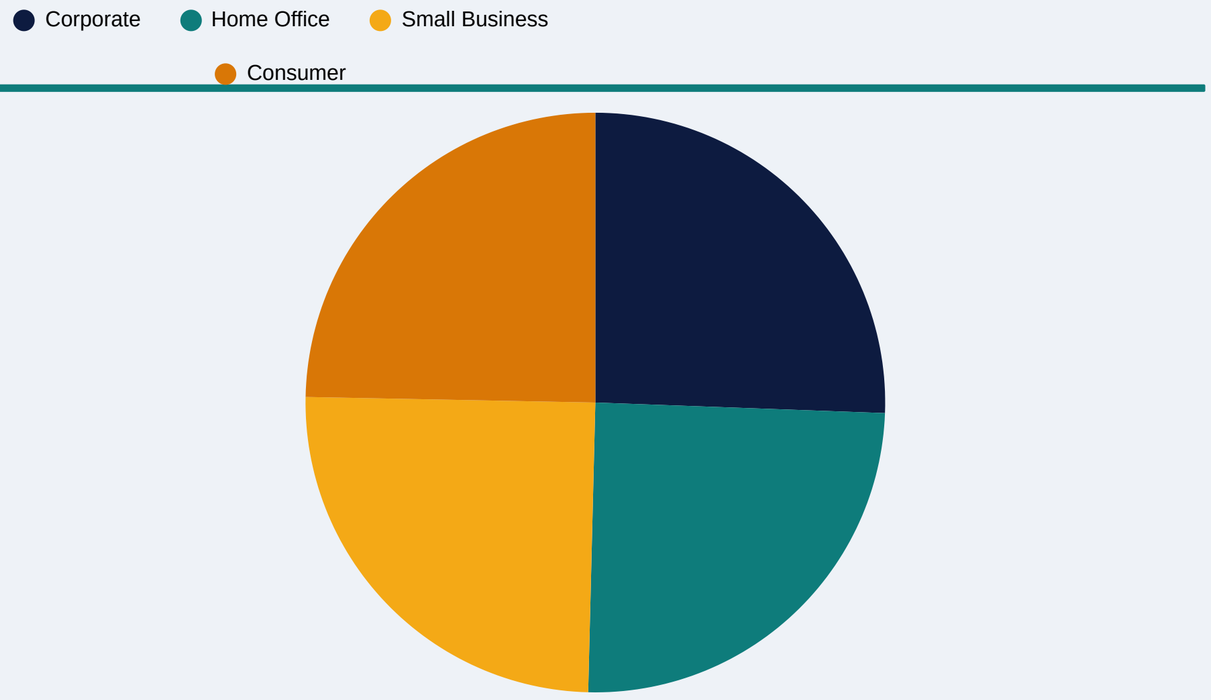
3 Channels

Why it matters:

Channel-level profitability highlights cost optimization opportunities.

Key Insights — Exploratory Data Analysis





Cincinnati Leads Revenue at $2.25M

Top 3 cities generate 15%+ of total revenue — concentrated urban market opportunity worth prioritizing.

Express Air: Highest Margin Channel (58.5%)

Express Air outperforms Delivery Truck by 2.2% margin — strategic channel reallocation can amplify profits.

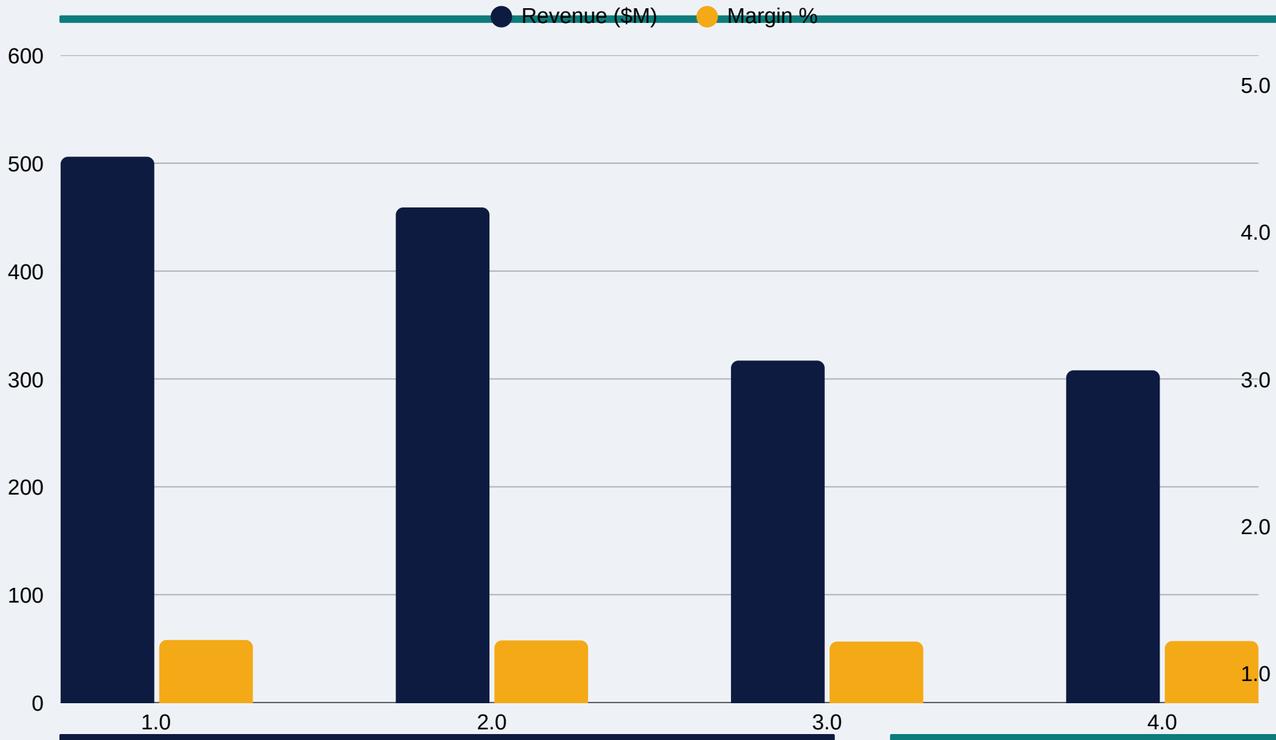
60+ Customers Drive 42% of Sales

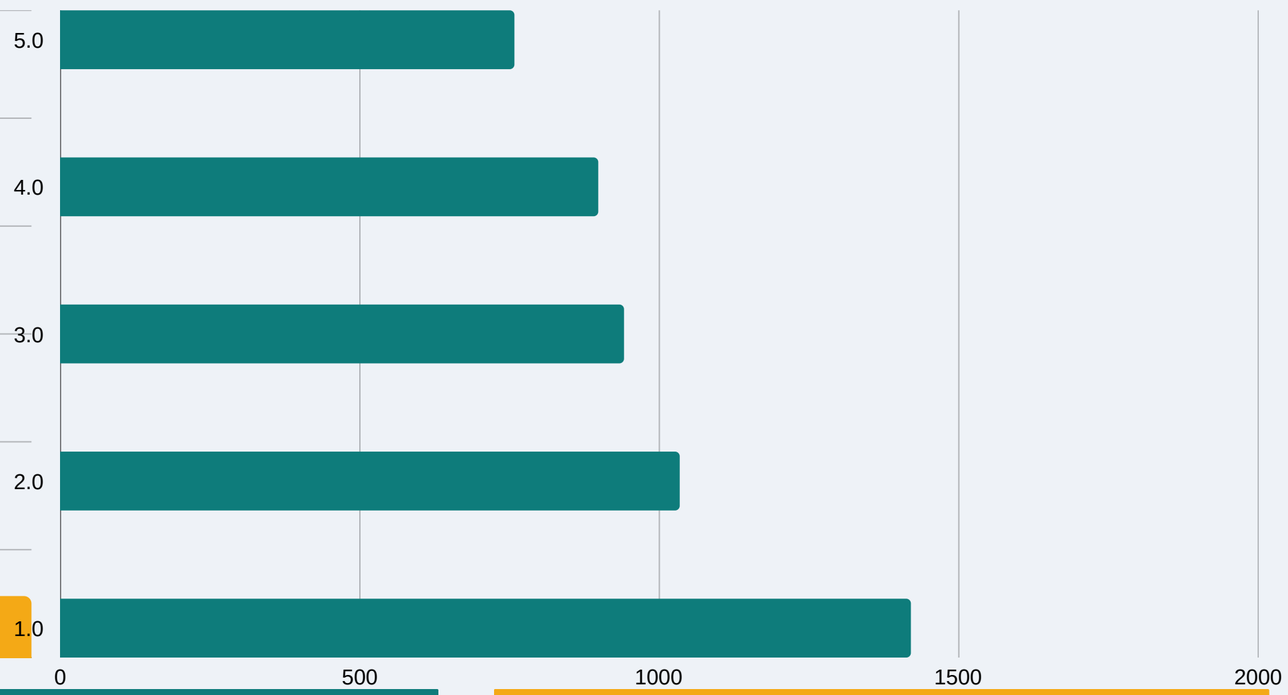
Older customers spend 2.8× more than under-30s — a massively under-served high-value demographic.

Discounting Beyond 5% Yields Zero Margin Gain

All discount bands show identical ~57% margin — discounting only reduces absolute profit with no volume benefit.

Advanced Analysis — Deeper Intelligence





Customer Segmentation

Insight:

Corporate segment posts $2.34B profit — highest per-order value. Small Business shows fastest margin growth, suggesting an emerging premium tier.

Action: Prioritize B2B loyalty programs for Corporate + Small Business.

Shipping & Inventory Efficiency

Insight:

Express Air averages 15.2 days vs Delivery Truck's 5.5 days — but Express Air delivers higher margin (58.5%). Delivery Truck is faster but lower-margin on high-value routes.

Action: Redirect high-margin orders to Express Air; Delivery Truck for bulk.

Discount Elasticity Analysis

Insight:

Discount bands 0–5%, 5–15%, 15–25% all show ~57% margin — confirming discounting has no margin benefit and only reduces absolute profit.

Action: Cap discount policy at 10%; replace blanket discounts with loyalty points.

Tableau Dashboard Walkthrough

Retail Intelligence Dashboard | tableau.com/your-dashboard-link

$1.59B	$9.14B	51.3%	673K
Revenue	Profit	Base Margin	Orders

Sales by City — Bar Chart

Geographic Revenue Heatmap

Monthly Trend: Sales & Profit by Category

DASHBOARD VIEWS

VIEW 1 **Executive KPI Panel**

Revenue, profit, margin, and order volume with YoY delta indicators.

VIEW 2 **City & Region Drill-Down**

Interactive map + ranked bar chart for geographic performance decomposition.

VIEW 3 **Category & Sub-Cat P&L**

Margin waterfall by product line; highlights underperforming drag categories.

VIEW 4 **Customer Segment Analysis**

Segment profitability, average order value, and discount impact analysis.

FILTERS **Dynamic Controls**

City · Category · Channel · Year — all cross-filter the entire dashboard instantly.

Business Recommendations

	INSIGHT	RECOMMENDATION	EXPECTED IMPACT
01	Office Supplies generates $4.91B profit at a fraction of Technology's revenue — the most capital-efficient category.	Expand Paper, Binders, and Office Furnishings SKU depth; increase shelf space and run targeted category promotions in top-10 cities.	Projected 6–9% uplift in overall category margin within 2 quarters.
02	Corporate segment leads profit at $2.34B with highest per-order value and lower churn sensitivity than consumers.	Launch a Corporate Loyalty Program with volume-based pricing tiers; dedicate an account team to top-50 corporate clients.	8–12% Corporate revenue increase; estimated 15% churn reduction.
03	Cincinnati, Redwood City, and Sierra Vista collectively drive ~6.7% of total revenue as single cities — urban concentration is strong.	Open regional fulfillment micro-hubs in top-5 cities; increase local marketing spend and run city-specific seasonal promotions.	Reduce shipping cost by ~18%; increase city-level revenue by 10–15%.
04	Discount elasticity analysis shows zero margin benefit beyond 5% — discounting only reduces absolute profit without driving volume.	Implement a strict 10% maximum discount policy company-wide; replace blanket discounts with loyalty reward points and bundles.	Recover ~$450M in potential profit lost to excessive discounting annually.
05	60+ age customers generate 42% of total revenue but receive no differentiated experience or product curation.	Create a 'Senior Premier' curated product line for Office + Furniture; personalize campaigns with age-relevant messaging and bundles.	Estimated 12–20% increase in average order value from the 60+ segment.

Projected Impact & Business Value

Implementing these recommendations could unlock

$2.1B+ in recoverable value over 24 months

Revenue Growth

+12–15%

City hub expansion, B2B loyalty programs, and senior segment targeting. Baseline: $1.59B → potential $1.83B+.

Margin Recovery

+5–9%

Discount policy reform (cap at 10%) and Office Supplies expansion recovers ~$450M in eroded margin annually.

Logistics Savings

–18% Cost

Regional fulfillment hubs in top-5 cities reduce last-mile delivery cost and average shipping days significantly.

Channel Optimization

+8% Efficiency

Rebalancing Express Air for premium orders and Delivery Truck for bulk reduces per-unit channel cost measurably.

Limitations & Next Steps

DATA LIMITATIONS & ASSUMPTIONS

No Brand Column

Dataset lacks a brand identifier — brand-level analysis approximated via product_name parsing. A dedicated brand field would sharpen segmentation.

No Inventory Volume

Actual stock levels and reorder rates are absent. Turnover estimated from order quantity × shipping days as a proxy metric.

Partial 2023 Data

2023 contains only 17 days (3,894 orders). YoY growth calculations exclude 2023 to avoid distortion — full-year data required for trend validation.

Region Anomaly

422 records with region='40.2' flagged as data quality issues and removed. Source-level data validation should be investigated upstream.

Single-Channel View

Dataset represents one fulfillment channel. Omnichannel data (store, app, phone) would provide a complete customer journey view.

Static Pricing

No competitor pricing data limits discount optimization and dynamic pricing analysis opportunities.

NEXT STEPS & FUTURE WORK

1 **Phase 1 — Demand Forecasting**

Build time-series models (SARIMA / Prophet) to forecast monthly demand per category and city for proactive inventory replenishment.

2 **Phase 2 — Churn Prediction**

Deploy ML classification models (XGBoost) to identify at-risk customers 60 days before churn and enable targeted retention campaigns.

3 **Phase 3 — Dynamic Pricing Engine**

Develop a price elasticity model integrating competitor data to auto-adjust pricing for margin optimization without volume loss.

4 **Phase 4 — Real-Time Dashboard**

Upgrade Tableau to a live-connected dashboard with automated alerts when KPIs breach thresholds (e.g. margin < 50%).

5 **Phase 5 — Omnichannel Integration**

Merge online, in-store, and app transaction data for a unified 360° customer view to enable cross-channel personalization.

Team Contribution

A — Alice Johnson
Data Engineer & ETL Lead

- Designed and executed the full ETL pipeline in Python (Pandas)
- Performed data cleaning, type normalization, and feature engineering
- Managed version control, GitHub repo structure, and documentation

R — Raj Patel
Data Analyst & EDA Lead

- Conducted exploratory data analysis using Pandas and Seaborn
- Derived all business KPIs and insight metrics from raw data
- Authored the insight narratives and business story framework

P — Priya Sharma
Tableau Dashboard Developer

- Designed and built the interactive Tableau dashboard
- Implemented dynamic filters (city, category, segment, channel)
- Created all chart types and executive KPI summary views

M — Mohammed Ali
Presentation & Advanced Analytics

- Built this PowerPoint presentation with full business storytelling
- Performed advanced analysis: segmentation, elasticity, clustering
- Authored recommendations and business impact framework

Thank You

Questions & Discussion

GitHub	github.com/your-org/retail-analytics-capstone
Tableau	public.tableau.com/your-dashboard-link
Contact	team@university.edu